

Article

The Effects of Integrating PBL Teaching Strategies with Two-Tier Mandala Thinking on Innovation Education

Yu-Chen Kuo *  and Shih-Ying Lee

Department of Computer Science and Information Management, Soochow University, Taipei 100006, Taiwan; alice20110315@gmail.com

* Correspondence: yckuo@scu.edu.tw

Abstract

In the digital era, industries increasingly demand innovation and problem-solving capabilities, making cross-disciplinary integration and creative thinking essential competencies for information management professionals. Although previous studies have shown that Problem-Based Learning (PBL) enhances students' problem-solving abilities and proactive learning behaviors, its effectiveness in supporting creative extension and conceptual deepening remains limited without structured thinking frameworks. To address this issue, this study integrated PBL with a Two-Tier Mandala Thinking approach based on a nine-grid structure. The proposed method combines first-tier divergent thinking with second-tier spiral convergence to guide students in establishing conceptual foundations, differentiating ideas, and refining design directions. A quasi-experimental study was conducted in a course in which students completed a game design task using either the Two-Tier Mandala Thinking Method or conventional brainstorming strategies. Quantitative results indicate that students in the Mandala Thinking group significantly outperformed those in the brainstorming group across three learning performance metrics. Qualitative findings further revealed that students using the proposed approach exhibited enhanced creative self-efficacy and greater confidence in their creative outcomes. Overall, integrating Two-Tier Mandala Thinking into PBL effectively supported the experimental group in structuring and developing in-depth creative thinking processes, providing empirical evidence for its application in innovation-oriented information education.

Keywords: PBL; Mandala Thinking Method; brainstorming; creative thinking; innovation education



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1. Introduction

1.1. Research Background and Motivation

Amidst the rapid transformations of the digital era, industries are increasingly demanding talent equipped with innovative capabilities and problem-solving skills [1]. These competencies are not isolated but develop sequentially and interdependently. Data from recent application exchange workshops and institutional feedback collected by the Ministry of Education's University Competency and Career Network (UCAN) platform indicate that students' growth in these two workplace competencies has been relatively slow or shows room for improvement. Consequently, these abilities have been prioritized for enhancement in campus teaching and competency development [2]. In the face of the rapidly changing social and industrial landscape of the 21st century, scholars emphasize

that advanced competencies such as creativity and critical thinking are crucial for students to navigate future uncertainties and achieve sustained development [3].

In current educational trends, creative thinking abilities are recognized globally as a critical element in talent development systems and a key indicator of learning effectiveness [4]. Optimized teaching processes and methodologies can significantly enhance students' creative thinking potential, which is not a static quality.

Specifically, innovative practices—such as producing original and creative works in project-based learning—can be regarded as manifestations of students' learning achievements resulting from the integration of professional skills and innovative thinking abilities [5].

The latest industry trends underscore the urgency of this issue. According to Cheers Magazine's 2024 survey report on "The Most Sought-After College Graduates by Companies", "problem-solving ability" has become the top trait employers value most. Additionally, the University Brand Power Report by 104 Job Bank emphasizes that companies highly value students' job competencies and integration of industry and academia, indicating that the industry expects students to effectively translate academic knowledge into practical applications [6,7]. These expectations suggest that basic problem-solving skills alone are insufficient to address contemporary challenges. Both creative thinking and problem-solving fall within the core domain of Higher-Order Thinking Skills (HOTS) [8]. A successful information-based creative product must transcend the framework of single technical functions, simultaneously considering unique innovative value and exceptional user experience. In response to this pedagogical gap, the present study proposes an innovative instructional model that integrates Problem-Based Learning (PBL) with a Two-Tier Mandala Thinking approach. Through a systematic framework, it guides students from logical analysis toward multi-perspective divergence and convergence. While maintaining rigor, it cultivates multi-level thinking and information creativity, ultimately producing mobile application designs that are both functional and original.

Within this instructional context, a game development environment (Unity) was introduced to ground students' design activities in a realistic scenario; however, the study's focus remained on fostering creative learning and conceptual design processes rather than technical development.

Information Management has long focused on information technology training, equipping students with the technical competencies required of engineers. However, training that focuses solely on technical skills is no longer sufficient to meet the industry's actual demand for talent. Research indicates that modern industries increasingly value individuals who can integrate knowledge across different fields and possess creative thinking, problem-solving, and collaboration skills to navigate complex and ever-changing work environments [9]. Research evidence suggests that applying the PBL model encourages inquiry-based learning through real-world problems and supports the development of collaboration and problem-solving skills [10,11]. Moreover, empirical studies in STEM education have consistently reported positive effects of PBL on students' creative thinking abilities [12].

Mandala Thinking, as a visual and structured thinking tool, has gained increasing attention in creative education over the past few years. Research integrating Mandala Thinking with multisensory design in culinary design courses has demonstrated that it significantly enhances students' creativity and aesthetic abilities [13]. Additionally, scholars have proposed the "Mandala of Creative Pedagogies" framework to explore the application of mandalas in the design of creative teaching in higher education and faculty professional development [14]. Collectively, these studies suggest that guiding students to think from multi-level perspectives through mandala structures facilitates both divergent and convergent thinking. Specifically, the two-tier Mandala adopted in this

research employs a deeper structural framework to promote the systematic decomposition and cross-level integration of complex problems in the information domain.

In this study, problem-based learning (PBL) serves as the common instructional approach for both the experimental and control groups, providing an authentic problem context for game design. The two groups differ only in the creative thinking tools employed during the learning process.

Prior studies have shown that Problem-Based Learning enhances creativity and collaboration. Other research highlights the potential of Mandala-based thinking as a visual tool that supports both divergent and convergent thinking. However, these two approaches have mostly been studied independently. Existing studies have rarely examined how integrating PBL with a multi-tier Mandala thinking framework works as a cognitive-process-oriented instructional design rather than as a general creativity technique. Furthermore, there is limited empirical evidence on the use of such integrated approaches in mobile technology or game design contexts, especially regarding students' creative learning outcomes, creative self-efficacy, and innovative behavior.

As a result, there remains a need for instructional designs that explicitly structure students' creative thinking processes within PBL environments, particularly in technology-oriented courses.

In summary, while both PBL and Mandala thinking demonstrate potential in theory and practice, few studies have combined them for application in mobile technology or information technology courses—particularly in game design contexts—to explore their impact on students' creative thinking and innovative behaviors.

1.2. Research Purpose

This study examined the effectiveness of integrating a Two-Tier Mandala Thinking Method into problem-based learning for game design. It compares different creative thinking tools in terms of learners' creative learning effectiveness, creative self-efficacy, and innovative behavior. To address this, the following research questions were proposed:

RQ1: Does the Two-Tier Mandala Thinking Method lead to significantly different creative learning effectiveness compared to conventional brainstorming?

RQ2: Does the Two-Tier Mandala Thinking Method lead to significantly different creative self-efficacy compared to conventional brainstorming?

RQ3: Does the Two-Tier Mandala Thinking Method lead to significantly different innovative behavior compared to conventional brainstorming?

2. Literature Review

2.1. Problem-Based Learning (PBL)

- The Origins and Significance of PBL Instruction

The Problem-Based Learning (PBL) instructional model traces its origins to the 1960s, when Professor Barrows developed it at McMaster University's Faculty of Medicine in Canada to enhance clinical teaching. Barrows observed that while traditional large-class lecture-based instruction effectively conveyed knowledge, it struggled to cultivate the integrated problem-solving and decision-making skills students needed to navigate the complex challenges of real clinical scenarios. To this day, the principles of PBL remain widely recognized as a means to address the limitations of traditional teaching models in cultivating higher order thinking and practical skills [15,16].

- Core Concepts of PBL Instruction

Contemporary educational research suggests that when students lack opportunities to solve real-world problems during the learning process, they fail to develop practical

problem-solving skills and have insufficient opportunities to explore and express their personal insights. To address this educational challenge, this study outlines the core characteristics of Problem-Based Learning (PBL) as follows:

I. Problem-Based Learning as the Core of Education:

The essence of PBL lies in placing a real, complex, and ill-structured problem at the heart of the learning process [17]. Such problems serve as driving forces, effectively stimulating students' motivation and deep thinking. A well-designed PBL problem should relate to students' life experiences or future professional fields, making the situation seem worthy of the time and effort to explore. In the PBL process, students first confront the problem scenario, then are guided to identify gaps in existing knowledge, and proactively gather and integrate cross-disciplinary information to construct a comprehensive and reasonable solution. Recent research confirms that this learning approach, oriented toward solving real-world challenges, not only significantly enhances students' problem-solving abilities but also promotes critical thinking and the practical application of knowledge [18]. This study's curriculum will incorporate current industry case studies, such as popular game design applications, as learning triggers. This approach guides students in connecting theoretical knowledge to real-world contexts, thereby deepening their learning.

II. Conducted in Group Collaboration:

Compared to traditional individual learning, PBL places a strong emphasis on the importance of "collaborative group learning". Students are not merely learners in the classroom but also partners in collaborative problem-solving. Within teams, each member contributes distinct perspectives, knowledge, and skills. Through frequent communication, discussion, negotiation, and knowledge sharing, they collectively construct an understanding of the problem and devise solutions [19]. This process not only aggregates diverse viewpoints but also prompts students to reflect on their own positions, thereby enhancing communication, collaboration, and critical thinking skills. Research by Dolmans et al. (2022) [19] indicates that the collaborative environment in PBL effectively cultivates students' teamwork spirit and interpersonal communication skills—core competencies indispensable in future workplaces.

III. Emphasizing Student-Centered Autonomous Learning:

The PBL teaching philosophy emphasizes placing students at the core of the learning process, gradually shifting the initiative from teachers to learners. In this student-centered model, teachers provide problem scenarios as the starting point for learning. Students must proactively analyze problems, define learning objectives, plan their progress, and independently search for, evaluate, and apply necessary information [20].

Related research indicates that PBL significantly enhances students' readiness for autonomous learning, transforming them from passive knowledge recipients into active knowledge constructors [20]. This approach encourages students to proactively explore the broad dimensions of "mobile technology and applications", making the learning process more meaningful and enjoyable.

IV. Teachers as Facilitators and Environment Builders:

In PBL classrooms, teachers transition from traditional knowledge disseminators to facilitators who guide learning. The core task of facilitators is not to provide direct answers, but to guide and deepen student thinking through questioning and creating a safe learning environment. Effective facilitators employ metacognitive questioning to challenge students' cognitive frameworks and help them overcome thinking bottlenecks. Simultaneously, facilitators maintain group momentum, ensure discussions stay on topic, and cultivate an environment where students feel empowered to ask questions and express divergent

viewpoints. A supportive learning atmosphere is crucial for PBL success [21]. The industry mentor in this study will adopt a highly guided facilitation strategy. By sharing personal experiences and posing probing questions, they will progressively stimulate students' creative thinking. The mentor will also strive to establish a respectful and open classroom environment, ensuring that every student's perspective is heard and valued.

Although PBL provides authentic problem contexts and supports collaboration and problem-solving, it does not explicitly specify how students' creative thinking processes are structured during ideation, leaving room for integrating structured creative thinking tools.

2.2. Mandala Thinking Method

- Origin and Definition

The term "Mandala" originates from Sanskrit and means "circle" or "attainment of essence". In Hinduism and Buddhism, practitioners initially used Mandala diagrams to symbolize the universe and guide meditation. Japanese scholar Hiroaki Imaizumi later adapted this concept and developed the Mandala Thinking Method, a systematic thinking tool centered on a nine-grid structure. This method borrows the "center-periphery" correspondence from mandala diagrams, placing the core theme at the center and using the surrounding eight squares for association and extension. Its purpose is to concretize and structure abstract thoughts, thereby stimulating deeper creativity and insight.

- Theoretical Basis of Divergent and Convergent Thinking

In creativity research, divergent and convergent thinking are widely recognized as two complementary cognitive processes involved in creative problem-solving [22,23]. Divergent thinking refers to the generation of multiple, diverse, and original ideas in response to an open-ended problem, whereas convergent thinking emphasizes the evaluation, organization, and refinement of ideas toward feasible and appropriate solutions. Although divergent thinking has traditionally received greater emphasis in creative research, convergent thinking also plays an equally important role in transforming ideas into coherent, workable outcomes [24].

These two processes are often considered interdependent and are commonly structured sequentially in creative frameworks, in which idea expansion precedes idea integration and selection. Such a divergent-to-convergent sequence is regarded as beneficial for avoiding premature fixation and for supporting a more systematic and effective creative process.

- Core Concepts and Thinking Patterns

The Mandala Thinking Method is a visualization tool based on a nine-grid structure. It guides learners in generating ideas and analyzing problems through a concrete grid framework. The process involves placing the core theme in the central grid and extending related ideas into the surrounding eight grids, enabling learners to examine the same issue from multiple perspectives. This structured, visually guided design supports the generation of diverse ideas and facilitates the creative thinking process [25]. Based on its operational approach across different learning tasks and application objectives, the Mandala Thinking Method manifests two primary cognitive orientations in its practical implementation: divergent thinking, centered on concept expansion, and spiral thinking, oriented toward organization, planning, and analysis. The following sections detail their respective operational methods and applicable scenarios:

I. Radiant Thinking:

This method places the core theme in the central cell, with the surrounding eight cells used to fill in keywords, concepts, or tasks directly associated with the center. There is no inherent sequence among the cells, encouraging free, divergent horizontal thinking. It is

suitable for the early stages of brainstorming as it rapidly expands the breadth of creativity. This method of generating visual concepts—characterized by outward expansion from a central point and emphasizing non-linear, multi-directional development—aligns with the divergent thinking process described in creativity research and is considered beneficial for expanding ideas during the early stages of the creative process [26], As shown in Figure 1.

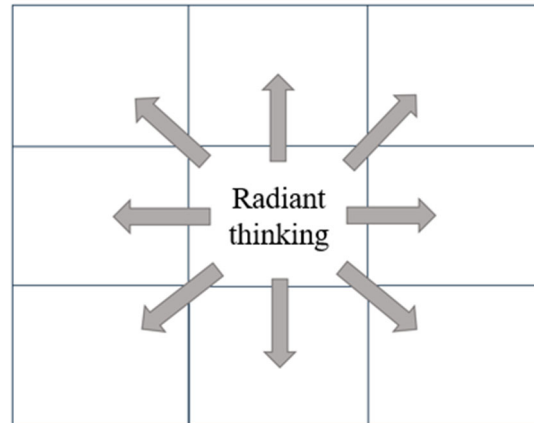


Figure 1. Radiant Thinking.

II. Spiral Thinking:

This method also originates from the central cell, but the surrounding eight cells follow a clockwise or counterclockwise direction, enabling vertical thinking with logical sequence or temporal progression. This approach emphasizes steps, processes, and causal relationships, making it suitable for project planning, strategy formulation, or in-depth problem analysis. This model resembles the structured knowledge organization and analytical activities referenced in educational research. Studies indicate that through visual structuring tools such as concept mapping, learners can more effectively arrange multiple concepts clearly and connect causal relationships and processes, thereby supporting systematic thinking and deep understanding [27], as shown in Figure 2.

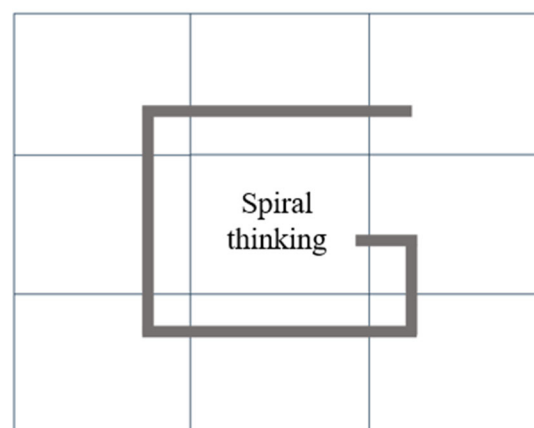


Figure 2. Spiral Thinking.

This study proposes employing a “Two-Tier Mandala Thinking Method” to guide students in the creative design of games. This approach integrates the strengths of radial and spiral thinking, forming a systematic design process that combines both divergent and convergent thinking. In the first tier, students use radial thinking to engage in open-ended brainstorming on the theme of “game design”. This process aims to explore diverse creative

possibilities and generate multifaceted conceptual elements, such as gameplay mechanics, world-building, and character traits.

Upon entering the second stage, researchers will provide specific constraints (e.g., target audience, technical limitations, narrative themes). Students must then employ the new mandala chart to conduct spiral thinking based on these parameters. Research indicates that introducing appropriate constraints during the later stages of creative ideation helps converge divergent ideas and generate more concrete, innovative solutions [28]. Through this two-tier design approach, students not only unleash their imagination but also learn to integrate and deepen fragmented ideas within real-world frameworks, ultimately developing a structurally sound and feasible design solution.

This Two-Tier Mandala structure functions as an explicit cognitive scaffold that guides both idea expansion and refinement within problem-based learning contexts.

2.3. Brainstorming

- Origins and Definition of Brainstorming

Advertising executive Alex Faickney Osborn first proposed brainstorming in 1939 and later systematically elaborated on the concept in his seminal 1953 work, *Applied Imagination*. This method aims to solve specific problems creatively through group collaboration. Contemporary research defines brainstorming as a structured process that facilitates divergent thinking within teams. Its core lies in generating a large volume of diverse ideas within a short timeframe by collectively sparking group wisdom [29]. The method's significance lies not merely in quantity but in fostering innovative concepts through the collision and inspiration of ideas among members—concepts often inaccessible through individual independent thinking. In the context of increasingly prevalent digital collaboration and remote work, brainstorming extends beyond physical meeting rooms. Instead, it is widely used through various online platforms and virtual environments for electronic or virtual brainstorming, serving as a vital tool that supports geographically dispersed members in collective creative thinking [30].

- Core Principles of Brainstorming

To ensure effective brainstorming, Osborn proposed four fundamental principles. These remain the cornerstone for guiding creative groups and continue to be validated in research today.

1. Delay Judgment:

This is the most critical principle. During idea generation, participants must not criticize or evaluate ideas. The purpose is to create a psychologically safe environment where everyone can propose even immature or wild ideas. This maximizes creative potential [31,32].

2. Welcome Free Association:

Encourage the proposal of any ideas that may seem absurd, impractical, or extreme. These “wild” ideas often break through existing mental frameworks, serving as catalysts for breakthrough solutions. Even if these ideas are not feasible on their own, they may inspire other members to generate more valuable connections [31].

3. Pursuit of Quantity:

This principle emphasizes that “quantitative change leads to qualitative change”. The more ideas a team generates, the higher the probability of producing high-quality, feasible solutions. A large volume of ideas also provides a rich repository for subsequent integration and screening [31].

4. Integration and Improvement:

Encourage careful listening and building on others' ideas through combination, modification, or extension. This technique is called "piggybacking". By recombining or refining concepts, teams can create solutions that are more innovative than any single idea.

While brainstorming is effective for generating a large number of ideas, it provides limited guidance for organizing and refining ideas during later stages of creative problem-solving.

2.4. Unity as a Conceptual Prototyping Tool

Unity is a cross-platform 2D and 3D game engine that is widely adopted in educational and creative contexts for rapid game prototyping. In this study, Unity's use was explicitly limited to presenting students' conceptual game designs, rather than serving as a focus for technical skill development.

Unity's visual interface and low technical entry barrier have made it popular in game design education, enabling learners without programming backgrounds to present conceptual ideas interactively [33].

In the present study, Unity functioned only as a development environment to facilitate students' presentation of their conceptual game designs. The core instructional emphasis remained on creative thinking and design, rather than on technical development or the quality of the final products.

Accordingly, Unity was never positioned as an instructional objective, analytical variable, or learning outcome in this research. It served solely as a neutral, contextual means for students to express and communicate their creative concepts during their final presentations.

3. Research Methods

3.1. Research Structure

This study encompasses the following three research variables, described as follows:

I. Independent Variable:

The independent variable had two groups, each defined by a teaching strategy. The experimental group used the Two-Tier Mandala Thinking Method. The control group used the General Brainstorming Method. This study examined how these two strategies affect learners and compared their results.

II. Dependent Variables:

These included creative learning outcomes, creative self-efficacy, and innovative behavior. After completing the learning process, learners completed questionnaires. Data were analyzed using homogeneity-of-slopes tests, Levene's tests for homogeneity of variance, and single-factor analysis of covariance (ANCOVA) to examine pre- and post-learning differences in creative learning outcomes, creative self-efficacy, and innovative behavior.

III. Control Variables:

To enhance internal validity and prevent extraneous variables from influencing the results, this study assigned all participants to the same instructor's Mobile Technology and Applications course. The curriculum content was identical across groups, ensuring learners possessed comparable prior knowledge levels.

3.2. Research Design

This study employed Problem-Based Learning (PBL) as a teaching strategy, integrated with a Two-Tier Mandala Thinking approach, to implement action-oriented innovation

design instruction. The aim was to investigate the impact of this innovative tool on learners' creative learning outcomes, creative self-efficacy, and innovative behaviors.

In this course, a game development environment (Unity) was introduced to provide students with a concrete design context for mobile game applications. Importantly, Unity was not treated as an instructional variable or a focus of technical skill training; rather, it served as a background instructional tool to support students' conceptual game design and creative ideation within the PBL framework.

During the course, students in the experimental group adopted a PBL approach combined with the Two-Tier Mandala Thinking Method. Guided by industry mentors, participants first engaged in PBL's problem-based teaching methodology, using real-world scenarios (such as game design) to stimulate critical thinking. To aid comprehension, mentors provided game-integrated SWOT analyses (as shown in Figure 3). Next, students entered the first tier. After placing the design theme at the center of the nine-grid layout, they freely generated creativity, filling in the eight required elements and related creative content to engage in divergent thinking and share diverse perspectives (as shown in Figure 4). Subsequently, participants moved to the second tier, where divergent thinking transitioned to constraint-based thinking (i.e., the spiral thinking method). Instructors provided specific constraints, such as conducting SWOT analysis or focusing on core themes (as shown in Figure 5), to help students clarify and refine their core design objectives. These constraints prevent ideas from becoming overly numerous or straying from the theme, thereby converging creative concepts into concrete and achievable design solutions.

Meanwhile, the control group students learn using PBL teaching strategies combined with conventional brainstorming methods. Students similarly selected case studies from examples provided by industry mentors for analysis but subsequently employed conventional brainstorming techniques to generate creative game ideas. Throughout the ideation process, the instructional design imposed no constraints. Instead, it encouraged students to think freely, fully utilize their imagination, and exchange ideas with group members without criticism. Guided by the instructor's business insights, students ultimately completed their game designs. Both groups then engaged in peer review and reflection through final presentations and mutual evaluations.

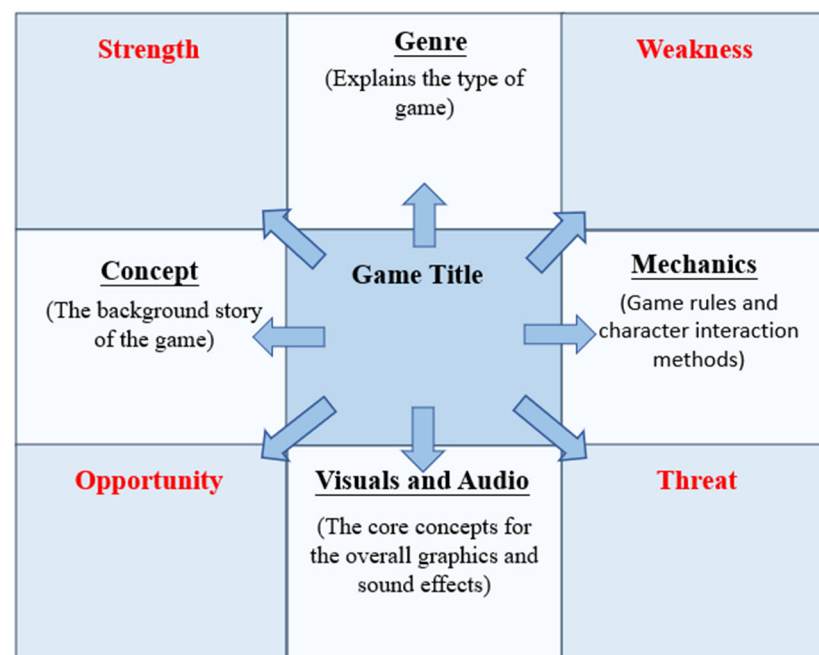


Figure 3. Two-Tier Mandala Thinking Combined with SWOT Analysis for Game Design.

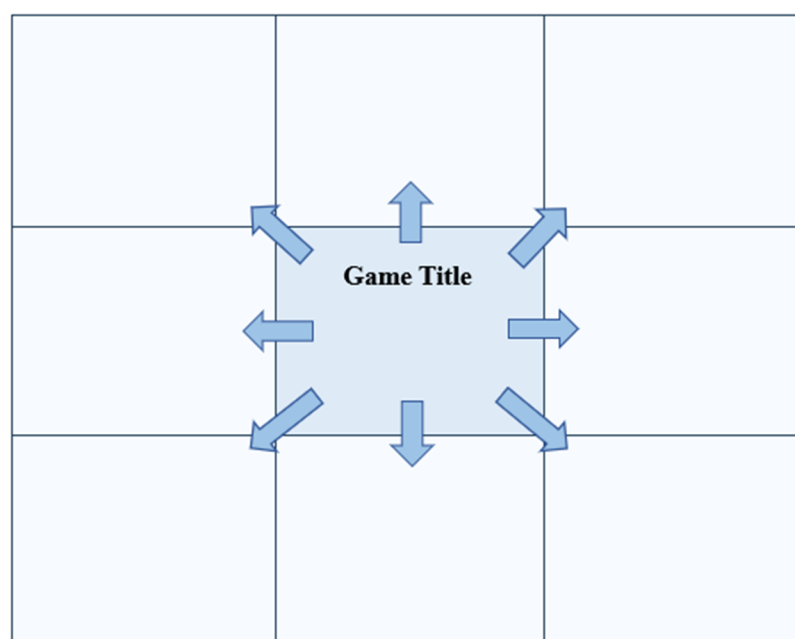


Figure 4. First-Tier Mandala Thinking Method—Divergent Thinking for Game Innovation Design.

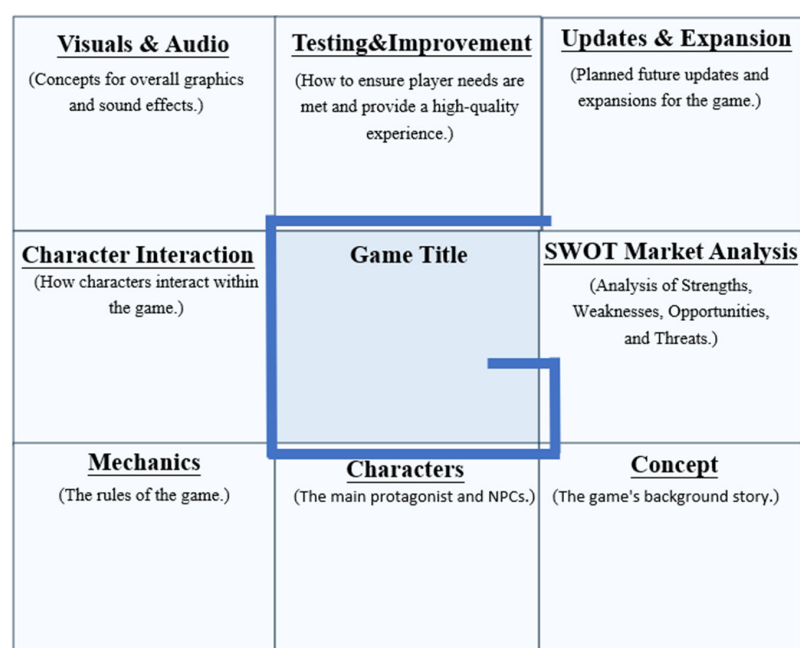


Figure 5. Second-Tier Mandala Thinking Method—Spiral Thinking for Game Innovation Design.

After both learning groups completed the instructional intervention, this study administered a post-test questionnaire. Using pre-test scores as a covariate, this study conducted a one-way analysis of covariance (ANCOVA) to examine the relationship between the two teaching strategies and learning outcomes. The study employed questionnaires to examine differences in students' creative learning outcomes, creative self-efficacy, and innovative behaviors. Ultimately, this research aimed to investigate whether the PBL teaching strategy, combined with the Two-Tier Mandala Thinking Method, enhanced the learners' creative learning outcomes, creative self-efficacy, and innovative behaviors more effectively than when combined with conventional brainstorming methods.

3.3. Experimental Design

3.3.1. Experimental Participants

The primary participants were students enrolled in the “Mobile Technology and Applications” course at the university’s Department of Information Management. This study divided 60 participants into an experimental group and a control group, with 30 students in each group. The control group employed PBL combined with conventional brainstorming techniques, while the experimental group utilized PBL integrated with a Two-Tiered Mandala Thinking approach.

3.3.2. Experimental Process

To implement PBL teaching combined with two different thinking approaches for game innovation design, pre-questionnaires were collected prior to the implementation. At the end of the term, this study administered post-questionnaires on creative learning outcomes, creative self-efficacy, and innovative behavior to collect data. After collecting all data, this study applied descriptive statistics for data analysis. The study then conducted a one-way analysis of covariance (ANCOVA) to compare the differences between the experimental and control groups in terms of creative learning effectiveness, creative self-efficacy, and innovative behavior.

As shown in Figure 6, this study’s experiment spanned six weeks, with each learning session lasting 150 min. Both the experimental and control groups were formed through voluntary grouping. The control group learnt using PBL teaching strategies combined with conventional brainstorming techniques. Students selected one game from the examples provided by industry mentors for analysis and evaluation. Subsequently, they employed conventional brainstorming techniques to generate creative ideas for the game. Throughout this process, the instructional design imposed no constraints and encouraged students to think freely, fully utilize their imagination, exchange ideas with group members, and refrain from criticizing others’ perspectives. The experimental group learnt using the PBL teaching strategy combined with the Two-Tier Mandala Thinking Method. Students selected a game from examples provided by the instructor for analysis and evaluation. However, the instructor first imposed specific thinking constraints, such as SWOT analysis and innovation design axis analysis. Subsequently, students employed the Two-Tier Mandala Thinking Method: the first tier involves divergent thinking, allowing students to brainstorm elements needed for innovative game design freely; the second tier shifts to spiral thinking, requiring students to generate innovative game design ideas within the imposed constraints. Both the experimental and control groups presented their outcomes through peer-to-peer evaluations during midterm and final assessments. This approach facilitates mutual learning, discussion, and high-level interaction among students.

3.3.3. Research Tools

This study employed a mixed-methods approach, utilizing both questionnaires and semi-structured interviews.

1. Creative Learning Effectiveness Questionnaire:

This study developed a 37-item questionnaire based on “Torrance Test of Creativity for Adults: Abridged Version” revised by [34], incorporating cognitive items adapted from the “Williams Creativity Test” revised by [35]. It employs a 7-point Likert scale, with a Cronbach’s alpha of 0.936 for both pre- and post-tests [34]. The questionnaire primarily examines differences in creative learning outcomes before and after the participants’ learning activities.

2. Creative Self-Efficacy Questionnaire:

The creative self-efficacy questionnaire used in this study adopted the revised version by [36] of the “Student Creative Self-Efficacy Scale” developed by [37]. This 12-item questionnaire employs a 7-point Likert scale, and both pre- and post-tests yielded Cronbach’s alpha values of 0.857 [36]. This instrument primarily examines differences in learners’ creative self-efficacy before and after the activity.

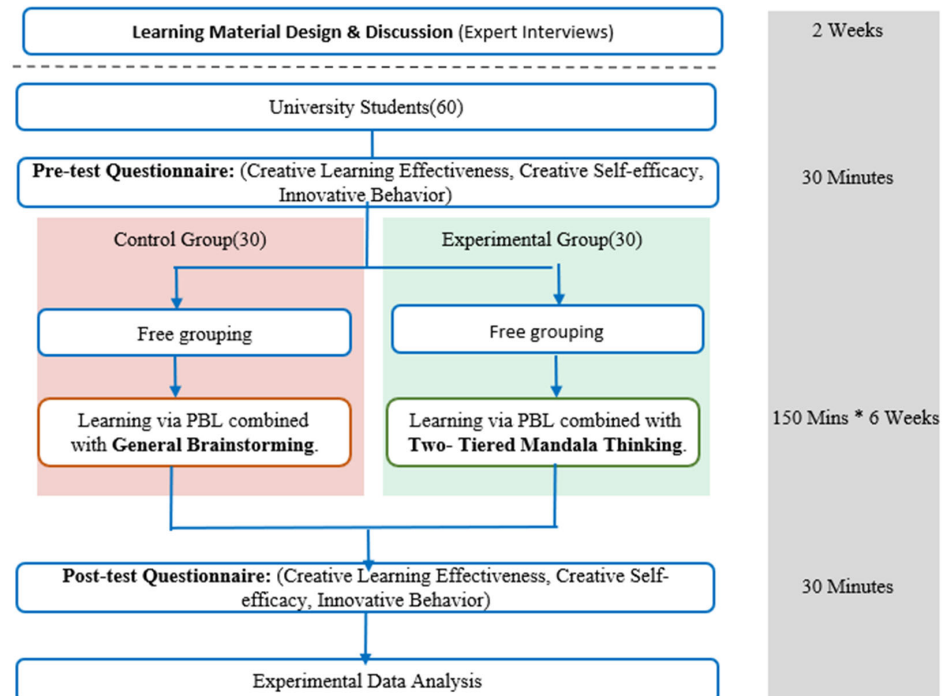


Figure 6. Experimental Flowchart. Note: * is a multiplication sign.

3. Innovation Behavior Questionnaire:

This study employed the Innovation Behavior Questionnaire, adapted from [38], which is a 7-item questionnaire that employs a 7-point Likert scale. The pre- and post-test Cronbach’s alpha coefficients were 0.86 [39]. It primarily examines differences in learners’ innovation behaviors before and after the activity.

4. Interview Items:

Adapted from the Creative Learning Effectiveness Questionnaire, Creative Self-Efficacy Questionnaire, and Innovation Behavior Questionnaire to gain detailed insights into the specific impacts on innovation and creativity-related dimensions. After the learning activities concluded, this study invited eight students from the experimental group (six females and two males) to participate in interviews to gain deeper insights into their perceptions of the learning activities and to conduct a qualitative analysis.

3.4. Data Analysis

Quantitative data collected from the pre-test and post-test questionnaires were analyzed using SPSS (version 22.0). Descriptive statistics were first conducted to summarize the participants’ performance in creative learning effectiveness, creative self-efficacy, and innovative behavior.

Given the quasi-experimental nature of the study, ANCOVA was employed to examine differences between the experimental and control groups while controlling for potential pre-existing differences. In each analysis, post-test scores served as the dependent variable, group (Two-Tier Mandala Thinking Method vs. conventional brainstorming) as the independent variable, and corresponding pre-test scores as covariates. Separate ANCOVA analyses were

conducted for each construct—specifically, each construction was analyzed individually to determine whether the intervention led to significant differences between groups for that construction. Before proceeding with these analyses, relevant assumptions were evaluated.

Before running ANCOVA, Levene’s test and regression slope homogeneity were checked and met.

4. Results

4.1. Questionnaire Analysis

4.1.1. Creative Learning Outcomes

Following the learning activities, this study administered a post-test to determine whether there were significant differences in creative learning outcomes between the “experimental group” and “control group” students. Using pre-test scores as a covariate, this study employed analysis of covariance (ANCOVA) to examine differences in creative learning abilities between the experimental and control groups. Levene’s test was performed before ANCOVA to assess the homogeneity of variances for post-test scores on creative learning effectiveness. Results showed that this assumption was not violated ($F = 0.939$, $p = 0.337$), confirming that ANCOVA was suitable. The analysis results, presented in Table 1, indicate that the post-test scores of the “experimental group” were significantly higher than those of the “control group” ($F = 14.796$, $p = 0.001 < 0.01$). The results suggest that learners in the experimental group, by adopting the Two-Tier Mandala Thinking Method, effectively enhanced their creative learning outcomes during the creative thinking processes facilitated by this method.

Table 1. ANCOVA test results for creative learning outcomes between experimental and control groups.

Group	N	Mean	Standard Deviation	Adjusted Mean	F	η^2
Experimental	30	5.372	0.847	5.291	14.796 **	0.206
Control	30	4.813	0.447	4.895		

** $p < 0.01$.

4.1.2. Creative Self-Efficacy

After the learning activities, both groups completed post-test questionnaires on creative self-efficacy. The analysis employed an analysis of covariance (ANCOVA) with pre-test creative self-efficacy scores as covariates. Before conducting ANCOVA, the assumption of homogeneity of regression slopes was examined. This was conducted by testing the interaction between group and pre-test scores of creative self-efficacy. The interaction effect was not statistically significant ($F = 2.974$, $p = 0.090$). This indicated that the assumption of homogeneity of regression slopes was satisfied. Results are presented in Table 2, showing that the experimental group demonstrated significantly higher creative self-efficacy than the control group ($F = 6.014$, $p = 0.017 < 0.05$). This finding suggests that learners in the experimental group developed greater confidence in their creative abilities during the creative thinking process by employing the Two-Tier Mandala Thinking Method. This approach effectively enhanced the creative self-efficacy of students in the experimental group.

Table 2. ANCOVA test results for creative self-efficacy between experimental and control groups after learning.

Group	N	Mean	Standard Deviation	Adjusted Mean	F	η^2
Experimental	30	5.022	0.963	4.961	6.014 *	0.095
Control	30	4.503	0.499	4.564		

* $p < 0.05$.

4.1.3. Innovation Behavior Questionnaire

Following the learning activities, both groups completed an innovation behavior questionnaire. Before conducting ANCOVA, the homogeneity of regression slopes assumption was examined by testing the interaction between group and the pre-test scores of innovative learning. The interaction effect was not statistically significant ($F = 0.830$, $p = 0.366$), which indicates that the assumption of homogeneity of regression slopes was satisfied. The analysis results employed an analysis of covariance (ANCOVA) with the pre-innovation questionnaire as a covariate. As shown in Table 3, the innovation behavior of the “experimental group” was significantly higher than that of the “control group” ($F = 4.827$, $p = 0.032 < 0.05$). The results suggest that learners in the experimental group were more willing to share creative ideas during the creative thinking process by employing the Two-Tier Mandala Thinking Method. They expressed an intention to apply this method to other conceptual tasks in the future, thereby enhancing the innovative behavior of students in the experimental group.

Table 3. ANCOVA test results for innovative behavior after learning between experimental and control groups.

Group	N	Mean	Standard Deviation	Adjusted Mean	F	η^2
Experimental	30	5.238	0.859	5.159	4.827 *	0.032
Control	30	4.705	0.638	4.74		

* $p < 0.05$.

4.1.4. Correlation Analysis of the Three Creative Dimensions

This section examines the relationships among the three creative dimensions measured in this study, namely creative learning outcomes, creative self-efficacy, and innovative behavior. Pearson’s correlation analysis was conducted using post-test questionnaire scores to explore whether these constructions were interrelated within the proposed instructional context.

The results, as shown in Table 4, indicate that all three creative dimensions were significantly and positively correlated with one another. Creative learning outcomes were strongly correlated with creative self-efficacy ($r = 0.827$, $p < 0.001$) and innovative behavior ($r = 0.802$, $p < 0.001$). In addition, creative self-efficacy also demonstrated a strong positive correlation with innovative behavior ($r = 0.812$, $p < 0.001$). These findings suggest that learners who achieved higher levels of creative learning outcomes also tended to exhibit stronger confidence in their creative abilities and more frequent innovation-oriented behaviors.

Table 4. Pearson correlation analysis of the three creative dimensions.

Group	1	2	3
1. Creative Learning Outcomes	1	0.827 **	0.802 **
2. Creative Self-Efficacy		1	0.812 **
3. Innovative Behavior			1

** $p < 0.01$ (two-tailed).

Overall, the results indicate that the three creative dimensions did not develop independently but rather showed a high degree of interconnectedness. Within the instructional design integrating Problem-Based Learning and the Two-Tier Mandala Thinking approach, improvements in one creative dimension were accompanied by corresponding enhancements in the others. This pattern reflects an integrated creative learning process, in which cognitive performance, creative self-belief, and innovative actions mutually reinforce one another.

4.2. Interview Analysis

To further explore the perceptions and thoughts of learners in the experimental group regarding the PBL combined with the Two-Tier Mandala Teaching Method, eight students (six females, two male) were invited to participate in semi-structured interviews after the experimental activities concluded. Table 5 presents the respondent data. This study used semi-structured interview questions as supplementary tools for data collection. During the interviews, the researchers asked learners about their perceptions of creative learning outcomes, creative self-efficacy, innovative behaviors, and their overall views on the teaching tool. Table 6 lists the interview questions.

Table 5. Information of interviewed students.

Pseudonym	Age	Gender	Interview Date	Group	Location
Participant A	21	Female	26 May 2025	Experimental Group	University Laboratory
Participant B	22	Female	26 May 2025		
Participant C	22	Female	26 May 2025		
Participant D	21	Female	26 May 2025		
Participant E	22	Female	27 May 2025		
Participant F	22	Female	29 May 2025		
Participant G	21	Male	29 May 2025		
Participant H	21	Male	6 June 2025		

This study employed semi-structured interviews to collect qualitative data, aiming to achieve two objectives: first, to understand the mechanism by which the Mandala Nine-Grid influences the creative process; and second, to capture the participants' subjective experiences and the essence of meaning when using the tool in depth.

To achieve this objective, this study employed grounded theory as its primary analytical framework and systematic procedure [40–43], ensuring rigorous theoretical construction. Meanwhile, this study supplemented the research process with a phenomenological perspective to deepen the understanding of the learners' subjective experiences and the construction of meaning [44]. The researchers employed thematic analysis procedures to systematically organize the data and summarize emerging themes, thereby capturing the multi-layered connotations of human experience in detail [45]. This integrated strategy aims to leverage the systematic nature of grounded theory to construct the theoretical framework while enriching its theoretical depth through the sensitivity of phenomenology.

Table 6. Interview questions.

No.	Questions
1	During this project using the Two-Tier Mandala Method, some students showed a significant increase in their creative thinking scores. Did you find it easier to come up with inspiration during this creative task? Why or why not?
2	Regarding items like "I can think of different ways to handle things," your score improved greatly. Which specific cell or direction in the Two-Tier Mandala helped you the most? Can you share the idea you had at that time?
3	In the questionnaire, you mentioned you are more confident in finding answers that others might not think of. Did the Mandala Method make you feel more confident during this creative task?
4	Is there a specific idea in your final work that made you think, "I am truly creative"? How did you come up with it?
5	During the design process, did you go back to review or modify your initial ideas?
6	Did the Mandala Method help you discover any problems or areas for improvement that you hadn't initially considered? How did you adjust during that process?

Table 6. *Cont.*

No.	Questions
7	Compared to previous class assignments, what felt different about adding the Mandala Method to this creative task?
8	While working on this task, was there a specific stage where you felt particularly engaged or thought, “This is actually quite fun”?
9	Did you try to actively share the creative ideas you came up with during this task with your classmates or teacher? What was the result?
10	If you wanted others to understand the creativity of this work, how would you introduce or promote it?
11	Overall, do you feel that using the Mandala Method in this creative task was helpful to you? Why?
12	Was there a specific moment during this task that left a deep impression on you or triggered an emotional response (e.g., excitement, feeling stuck, being inspired)? Please share.

4.2.1. Steps for Analyzing Interview Data

Interview data analysis followed the systematic coding procedure of grounded theory, incorporating a phenomenological perspective at each stage:

(1) Transcription

After transcribing the interview recordings, the researchers conducted multiple readings of the transcripts. This stage aimed to minimize the influence of preconceived notions and quantitative outcomes, allowing the researchers to immerse themselves fully in the data. The analysis focused on the participants’ raw descriptions and inner feelings regarding the Nine-Grid experience to capture the essential meaning of their experiences.

(2) Interview Content Coding

Next, we proceeded to a systematic coding process, meticulously unraveling insights from vast, fragmented data.

- **Open Coding:** Break down interview content into initial concepts. At this stage, specifically flag any statements expressing emotions, subjective experiences, or cognitive judgments.
- **Main Axis Coding:** Using constant comparison, systematically compare the similarities and differences across the interviewees’ experiences to identify patterns and themes. Group initial concepts into abstract subcategories and explore relationships among these subcategories—such as conditions, phenomena, strategies, and outcomes. This stage aims to uncover the internal mechanisms within the creative process.

(3) Core Theory Construction and Theme Development (Selective Coding)

Finally, selective coding established interpretive core categories. All subcategories were systematically linked to these core categories to construct an explanatory theoretical model. Ultimately, the analysis organized these core theoretical concepts into clear themes to describe and explain the mechanisms through which the Nine-Grid influences the creative process, thereby facilitating interpretation of the deeper meanings embedded in the texts.

4.2.2. Transcription of Interview Transcripts

This interview involved eight participants. To ensure anonymity, the study conducted the interviews anonymously and coded participants as A, B, and so on. This step consisted of transcribing the participants’ interview content verbatim in text format, preserving the original content entirely to serve as the textual source for subsequent analysis.

4.2.3. Open Coding

All interviewee transcripts underwent content coding using the following methodology (Table 7). English codes represent distinct interviewees; for example, C-1-1 denotes the first paragraph of Interviewee C's response to the first question.

Table 7. Explanation of code meanings.

Coding Meaning		
Character One	Character Two	Character Three
Interviewee Number	Question Number	Response Segment
A~H	01–12	1–5

This phase condenses the verbatim transcripts of respondents' answers to distill their true meaning—that is, conceptualizing the content. Table 8 is an example of transcript coding for the interviews (partial excerpt only).

Table 8. Open coding examples.

Code	Conversation Content	Coding Significance
A-01-01	- Interviewer: "Since we used the Two-Tier Mandala Thinking Method this time, did you feel that inspiration appeared more easily during the creative process?" - Interviewee: "Yes. Because it provides us with a thinking context, we can follow this framework to think about exactly what kind of effect we want to achieve. A-01-01"	Confirms the appearance of inspiration; provides a thinking context and framework; guides expected outcomes.

4.2.4. Core Coding

The open coding from the previous step was further refined into more essential core concepts to form the core coding. Below are the five final core codes identified.

I. Thinking Structure and Guidance

Definition 1. *As a framework, the Mandala Thinking Method provides systematic support for project initiation, directional positioning, organization, and management, addressing the challenge respondents face in not knowing where to begin during the initial ideation phase.*

II. Creative Inspiration and Ideation

Definition 2. *The multidimensional prompts and divergent process of the Mandala Method effectively inspired respondents to generate unique, unexpected ideas and expanded the scope of their thinking.*

III. Reflection, Review, and Refinement

Definition 3. *Through the review points provided by the Mandala Method and continuous implementation feedback, respondents can identify deficiencies, omissions, or implementation challenges in their work. They then iterate and refine the project to enhance its quality and functionality.*

IV. Collaboration and Communication

Definition 4. *As a structured discussion tool, Mandala Thinking effectively enhances communication efficiency within teams, facilitates consensus building, and promotes the exchange and application of external perspectives.*

V. Creative Self-Efficacy and Confidence

Definition 5. *Through structured processes and the experience of successfully bringing creative ideas to fruition, respondents develop a strong sense of control, engagement, and accomplishment regarding their creative abilities and the quality of their output.*

4.2.5. Selective Coding

The final stage of the selective coding analysis process aims to systematically organize the categories and hierarchical relationships established through axial coding. By examining the essential connections between categories, this stage facilitates the conceptualization process. Table 9 presents the results of selective coding.

Table 9. Selective coding results.

Core Category	Primary Axial Coding	Basis for Classification & Logic
Creative Thinking Process	I. Thinking Structure & Guidance	Focuses on initiating and organizing thoughts. The Mandala Method provides a “contextual framework” and “conceptual grid” that solves the problem of not knowing where to start, serving as cognitive preparation and orientation.
	II. Creative Inspiration & Ideation	Focuses on generating and expanding ideas. Includes using “prompts” to generate unexpected items and using rhyming dictionaries or sudden creative flashes. This is the core generation phase.
Creative Outcome Transformation	III. Reflection, Review & Refinement	Focuses on quality control and turning ideas into feasible plans. Includes scaling down projects when encountering technical issues, discovering the need for SWOT analysis, and correcting overly complex designs. This is the iterative optimization phase.
	IV. Collaboration & Communication	Focuses on the execution mechanism of team ideas. Streamlines discussions for after-hours work, speeds up division of labor through “ordered processes,” and resolves disagreements by producing “compromise solutions”.
	VI. Creative Self-efficacy & Confidence	Focuses on the final psychological outcome. When interviewees see their ideas become “creative realities” and feel “great” about finishing the whole project, this sense of achievement serves as an indicator of successful transformation.

Based on the analysis of the aforementioned data, two core categories emerged from the interview content.

The creative thinking process comprises two main categories: “Thought Structure Guidance” and “Creative Inspiration and Ideation”, both of which pertain to how learners’ creative ideation processes evolve after applying the Mandala Thinking Method. The innovative outcome transformation process, meanwhile, encompasses “Reflection, Review, and Refinement”, “Collaboration and Communication”, and “Creative Self-Efficacy and Confidence”. These represent how learners implement their ideas into practice while building self-assurance and adaptability throughout the process.

Analysis of the above coding results revealed the following key points:

1. The Mandala Thinking Method helped learners clarify direction and categorization, making their thought processes more structured and efficient.
2. Learners generate more creative ideas, reducing mental blocks and anxiety.

3. Enhanced creative self-efficacy increases learners' confidence in accomplishing creative tasks.
4. Mandala facilitated faster integration of team members' ideas and smooth communication during consolidation.
5. Works created using Mandala encouraged most students to share their creations with others willingly.
6. The majority of students applied their creative ideas to designs and final products, demonstrating innovative behavior and practical skills.

5. Discussion

This study aimed to investigate whether integrating Problem-Based Learning (PBL) with the Two-Tier Mandala Thinking Method (experimental group) in the "Mobile Technology and Applications" course could more effectively enhance the students' creative learning outcomes, creative self-efficacy, and innovative behaviors compared to integrating conventional brainstorming techniques (control group). Based on the analysis of quantitative questionnaires and qualitative interviews, this study yielded the following significant empirical findings, substantiating the benefits of the proposed innovative teaching model:

RQ1: Differences in Creative Learning Outcomes

Quantitative results indicate that the experimental group achieved significantly higher creative learning outcomes after the intervention than the control group. This finding suggests that the Two-Tier Mandala Thinking Method effectively enhances students' creative learning abilities.

Qualitative interview findings confirmed that the Mandala Method provides a clear "thinking structure and guidance", offering students a "contextual framework for thought". These features help students overcome the common challenge of feeling "stuck" during the initial ideation phase. Such a structured tool makes the thinking process more systematic and efficient, facilitating creative inspiration and ideation, thereby making it easier for ideas to emerge. Several students explicitly stated in interviews that this structured guidance approach helped them gradually develop their ideas and clarified their thinking.

In other words, while traditional brainstorming emphasizes divergence, its lack of structure often leads to disorganized ideas. The Two-Tier Mandala effectively transforms creative concepts into concrete design solutions through its first tier of radial divergence, which expands the breadth of ideas, and its second tier of spiral convergence with constraints, which focuses depth and feasibility. This structured process enhances the quality and feasibility of ideas, ultimately reflected in higher scores on creative learning achievements.

This finding is consistent with [14], who proposed that mandala-based creative pedagogies provide structured visual frameworks that support both divergent and convergent thinking in higher education. In addition, this result aligns with the creativity framework proposed by [46], which emphasizes the integration of divergent idea generation and convergent refinement as a key mechanism for enhancing creative performance.

RQ2: Differences in Creative Self-Efficacy

Quantitative results indicate that the experimental group demonstrated significantly higher creative self-efficacy than the control group, suggesting that the Two-Tier Mandala Thinking Method effectively enhances students' confidence in their creative abilities.

The "Creative Self-Efficacy and Confidence" theme in qualitative interviews emphasized that this confidence boost stems from a "sense of control" and "sense of accomplishment" over the process. Some interviewed students noted that clear brainstorming steps and structured planning could reduce uncertainty about the creative process, thereby enhancing engagement and confidence.

The Mandala Thinking Method enabled students to “grasp the process and thematic direction”, clarifying their thought processes. More crucially, through this structured tool, students successfully transformed abstract ideas into concrete works of art. Students exceeded the prior expectations of their own abilities by successfully realizing creative concepts, demonstrating that they could implement ideas and meet functional requirements. Consequently, this experience substantially strengthened their confidence in both their capabilities and the quality of their output.

These results align with prior research suggesting that creative self-efficacy is strengthened when learners engage in structured, design-oriented learning processes that support iterative idea development and successful task completion, thereby enhancing their perceived control and confidence in creative problem solving [47].

RQ3: Differences in Innovative Behavior

Quantitative results indicate that the experimental group exhibited significantly higher levels of innovative behavior than the control group. These results suggest that students in the experimental group were more willing to share creative ideas and apply this approach to future conceptualization. Innovative behavior involves not only generating ideas but also implementing and sharing them.

Qualitative interviews revealed that the enhancement in implementation capability (innovative behavior) hinged on the Mandala Method’s second tier, which emphasizes incorporating constraints and SWOT analysis. These results force students to consider market strengths, weaknesses, opportunities, and threats. During the interviews, some students also indicated that this analytical process helped them re-examine their original ideas and uncover previously unnoticed issues and avenues for improvement. These results facilitated reflection, review, and refinement, enabling students to identify previously overlooked analytical dimensions, address implementation challenges, and shift their focus from superficial design to functionality and practicality. These results successfully transformed abstract ideas into viable, innovative outcomes.

The enhancement in sharing and application (innovative behavior) stems from collaboration and communication. As a shared visual tool, the Mandala Method makes ideas immediately clear, accelerates structured discussions to facilitate team division of labor, and improves communication and expression. These results help teams resolve disagreements. This efficient communication environment encourages most students to willingly share their creative work, demonstrating proactive, innovative behavior.

In summary, the PBL teaching strategy, combined with the Two-Tier Mandala Thinking Method provides a structured, multi-level, and comprehensive thinking framework that integrates both divergent and convergent thinking in information application courses. It not only addresses the limitations of traditional brainstorming methods in terms of systematicity and outcome translation but also, through successful implementation, significantly enhances students’ creative self-efficacy. These results empower them to transition from passive recipients to confident innovators equipped with practical capabilities.

This finding extends existing research on innovative behavior in educational contexts by demonstrating that structured creative thinking frameworks, particularly those that integrate analytical constraints and collaborative reflection, can facilitate not only idea generation but also idea implementation and knowledge sharing. Such results are consistent with prior studies highlighting the role of design-oriented and collaborative learning approaches in promoting students’ innovative behaviors and practical problem-solving capabilities [48].

Although creative learning outcomes, creative self-efficacy, and innovative behavior were all examined as outcome variables, they should not be interpreted as parallel outcomes. Creative learning outcomes reflect cognitive performance, creative self-efficacy reflects

creative confidence, and innovative behavior reflects the enactment of creativity. These constructs comprise different but interrelated layers of creative development.

The correlation analysis shows that these dimensions are strongly connected and support each other. They do not develop independently. In the instructional design, the Two-Tier Mandala Thinking Method and PBL affect each construct through distinct mechanisms. This supports a unified creative learning process rather than three separate effects.

6. Conclusions

Previous studies have shown that Problem-Based Learning (PBL) enhances students' problem-solving abilities and proactive learning behaviors. However, its effectiveness in supporting creative extension and conceptual deepening remains limited without structured thinking frameworks. To address this issue, this study integrated PBL with a Two-Tier Mandala Thinking approach based on a nine-grid structure. The proposed method combines first-tier divergent thinking with second-tier spiral convergence to guide students in establishing conceptual foundations, differentiating ideas, and refining design directions. A quasi-experimental study was conducted in a "Mobile Technology and Applications" course in which students completed a game design task using either the Two-Tier Mandala Thinking Method or conventional brainstorming strategies. Quantitative results indicate that students in the Mandala Thinking group significantly outperformed those in the brainstorming group across three learning performance metrics. Qualitative findings further revealed that students using the proposed approach exhibited enhanced creative self-efficacy and greater confidence in their creative outcomes. Overall, integrating Two-Tier Mandala Thinking into PBL effectively supported the experimental group in structuring and developing in-depth creative thinking processes, providing empirical evidence for its application in innovation-oriented information education.

6.1. Research Limitations

The sample for this study comprised 60 students enrolled in the Mobile Technology and Applications course within the Information Management Department of a university, with 30 participants in each of the experimental and control groups. As the experiment was conducted exclusively with students in this course, the sample size was relatively small. Consequently, the research findings are primarily applicable to learners with characteristics similar to those in this study. Future research should examine whether the proposed instructional approach can be generalized to students in other academic disciplines or learning contexts.

In addition, the instructional intervention was implemented within a limited course duration. As a result, the long-term effects of integrating the Problem-Based Learning with the Two-Tier Mandala Thinking Method on students' creative learning and innovative behaviors could not be fully examined. Future studies may consider longitudinal designs to further explore the sustainability of these learning outcomes.

Furthermore, this study intentionally focused on instructional innovation and students' creative thinking processes rather than on technical skill acquisition. Although Unity-based game development was incorporated into the course activities to provide an authentic design context, the Unity platform itself was not treated as a primary research variable or assessment focus. Therefore, the findings mainly reflect the effects of the proposed pedagogical approach and thinking framework on students' creative learning and innovation-related outcomes. Caution should be exercised when generalizing these results to learning environments involving different development platforms or technological contexts.

6.2. Future Research Directions

The PBL teaching method developed in this study, integrated with the Two-Tier Mandala Thinking approach, demonstrated that learners in the experimental group achieved significantly higher learning outcomes than those in the control group during the learning process of action technology and application courses. The learning effectiveness analysis in this study utilized pre- and post-test scores as quantitative metrics and conducted semi-structured interviews with students to capture detailed, concrete perceptions.

Therefore, future research should incorporate diverse qualitative data collection and analysis into the teaching process. Such approaches may include observing and analyzing learners' notes, visual ideation processes, and worksheet completion when they use mandala diagrams for topic decomposition and extension. Such approaches would further explore whether structured visual thinking aids students in clarifying concepts, enhancing comprehension, and correcting misconceptions.

Furthermore, subsequent stages of this study revealed that while some students could generate ideas using the Mandala Thinking Method, they struggled to assess the creativity and quality of their own outputs, lacking clear evaluation criteria. These findings suggest that simple idea-generation guidance can stimulate creative output but remains insufficient for fostering deep creativity and discerning quality. Future research may consider integrating the SCAMPER strategy-based "SCAMPER Method" to guide students in refining their work and reimagining creativity from multiple angles: Substitute, Combine, Adapt, Put to Other Uses, Eliminate, and Rearrange. The specific prompts of the SCAMPER method can encourage students to conduct critical analysis and self-assessment of their work, thereby strengthening their creative reflection and innovation capabilities. It also helps address the need for students to optimize and position their creative outcomes after the Mandala brainstorming process, enhancing overall teaching effectiveness and students' creative literacy.

In addition, future research may extend the present study by examining how different design or development environments shape students' creative learning processes when using structured visual thinking tools. Although Unity was used as the instructional context in this study, comparative investigations across different platforms or design environments may provide further insights into how technological contexts interact with PBL and Mandala-based thinking approaches to influence creativity and innovation-oriented learning outcomes.

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